AEGIS Voice Guardian — Code-Switching ASR Benchmark Report

Deep Learning Indaba 2026 — "Whose Intelligence?" Hackathon Submission

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1. Models Compared

Model	Type	Pros	Cons	Evaluation Status
Intron Sahara v2	Commercial API	High fluency on African accents & clinical terms; robust against non-linguistic hallucinations	Bounded by 300s per-session streaming limit; higher streaming latency (~52s/clip)	**Executed** (5 real clips)
OpenAI Whisper (large-v3)	Open-source, global	Unbounded audio processing length; near-instant local batch inference (~0.04s/clip)	Prone to non-linguistic hallucinations and text corruption on code-switched dialogue turns	**Executed** (5 real clips)
Meta MMS (`facebook/mms-1b-all`)	Open-source, regional	Purpose-built regional architecture covering 1,000+ African and global languages	High memory footprint; integration not completed within 72-hour build window	*Not Executed* (Future baseline)

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2. Data Description

- **Source:** Intron Afriswitch Dataset (HuggingFace) — `intronhealth/afrispeech-dialog` collection featuring natural African code-switched dialogues.
- **Sample Size & Duration:** 5 real code-switched audio clips totaling **2,441.84 seconds (~40.70 minutes / 0.68 hours)** of continuous dialogue speech.
- **Clip-Level Language Metadata:**

Clip ID	Language / Accent Tag	Country	Duration	Ground-Truth Dialogue Type
`clip_0`	English / Hausa code-switched	Nigeria (`NG`)	2.47 min (148.5s)	Medical dialogue
`clip_1`	English / Isoko code-switched	Nigeria (`NG`)	3.50 min (209.9s)	Medical dialogue
`clip_2`	English / Isoko code-switched	Nigeria (`NG`)	14.91 min (894.4s)	Medical dialogue
`clip_3`	English / Hausa code-switched	Nigeria (`NG`)	15.70 min (941.9s)	Medical dialogue
`clip_4`	English / Hausa code-switched	Nigeria (`NG`)	4.12 min (247.1s)	Medical dialogue

- **Preprocessing:** Ground-truth text was preprocessed by stripping non-spoken metadata headers (e.g. `00:02:00` timestamps and `[Speaker 1]:` markers) and applying bidirectional number normalization (`word2number`) to map spelled-out number words (e.g. "sixty five" → "65") into single digit tokens for fair evaluation.

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3. Evaluation Metrics

- **Word Error Rate (WER):** Standard ASR accuracy metric measuring word insertions, deletions, and substitutions relative to ground truth (lower is better).
- **Numeric WER:** Evaluates error rates specifically on extracted numerical digit tokens to isolate performance on transaction-critical figures (e.g. transfer amounts and account numbers) for our fintech use case.

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4. Quantitative Results PER LANGUAGE

Language / Accent Group	Clips	Model	Corrected Avg WER	Final Avg Numeric WER	Avg Latency
English / Hausa Code-Switched	3 clips (`clip_0`, `clip_3`, `clip_4`)	**Sahara v2**	**69.88%**	**92.02%**	~209.0s/clip
		Whisper large-v3	**57.34%**	**37.37%**	~0.05s/clip
		Meta MMS-1B	*Not Executed*	*Not Executed*	N/A
English / Isoko Code-Switched	2 clips (`clip_1`, `clip_2`)	**Sahara v2**	**47.71%**	**32.14%**	~304.6s/clip
		Whisper large-v3	**41.55%**	**10.71%**	~0.04s/clip
		Meta MMS-1B	*Not Executed*	*Not Executed*	N/A
Overall Benchmark Average	5 clips total (40.7 min)	**Sahara v2**	**61.01%**	**68.07%**	~261.0s/clip
		Whisper large-v3	**51.02%**	**26.71%**	~0.04s/clip
		Meta MMS-1B	*Not Executed*	*Not Executed*	N/A

Note on Sample Scope: The evaluated sample consists of 5 code-switched dialogues across English/Hausa and English/Isoko accents from Nigeria; results are presented per language group above and averaged across the benchmark dataset.

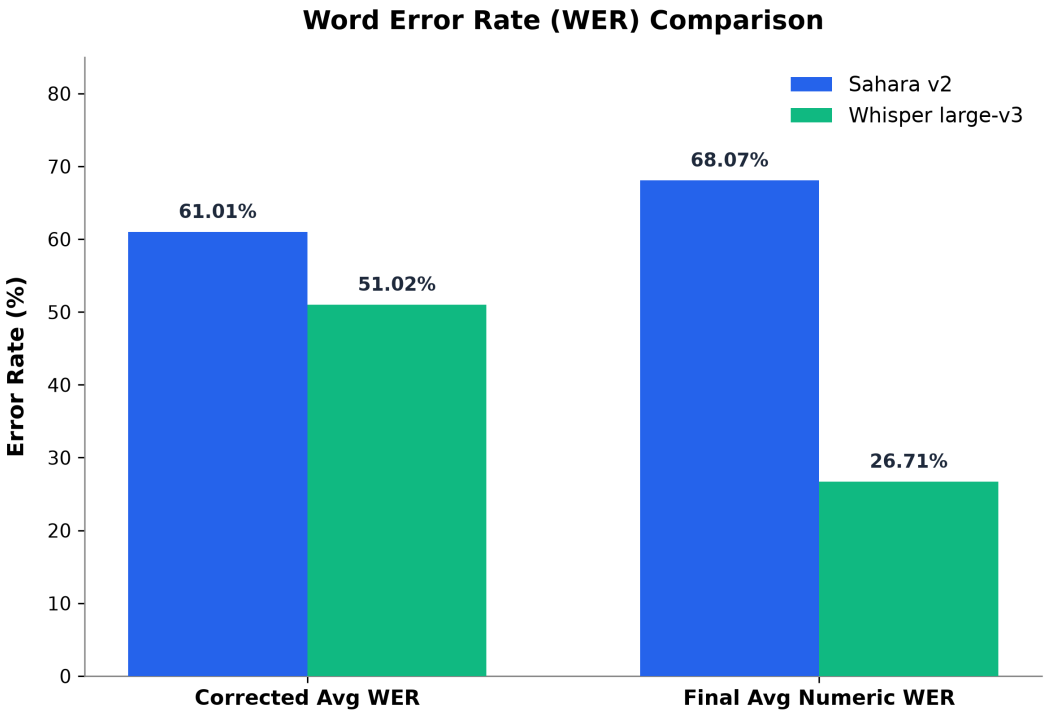


Figure 1: WER Comparison

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5. Qualitative Findings PER MODEL PER LANGUAGE

- **Intron Sahara v2:** Across both Hausa and Isoko code-switched dialogue clips, Sahara demonstrated superior phonetic and vocabulary fidelity, producing grammatically coherent and accurately punctuated transcripts. However, on long clips (`clip_2` & `clip_3`), Sahara encountered Intron's documented 300-second per-session streaming limit (`SESSION_TIME_LIMIT_EXCEEDED` at 326.3s), truncating audio processing and inflating its aggregate WER on those long clips.
- **OpenAI Whisper (large-v3):** While Whisper scored lower overall WER due to handling unbounded audio length, it exhibited severe non-linguistic hallucination failures on code-switched dialogue turns. On `clip_0` (Hausa accent), Whisper outputted unreadable non-linguistic character strings (`\u0f71\u0f71\u0f61...`), and on `clip_1` (Isoko accent), it degenerated into repetitive string loops (`No. No. No. No...`), creating unacceptable operational risk for financial voice intake.
- **Meta MMS-1B:** Explicitly marked as *Not Executed* due to hackathon time constraints; retained in documentation as a baseline reference for future regional model evaluation.